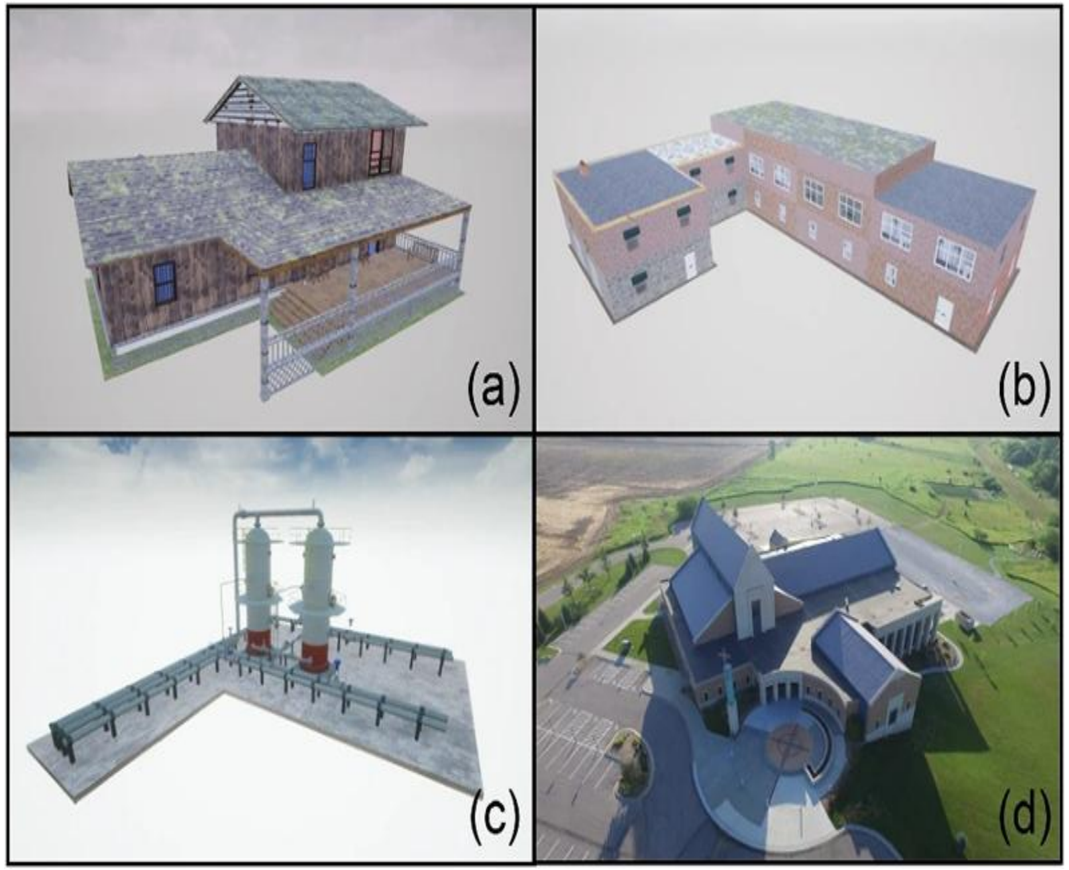
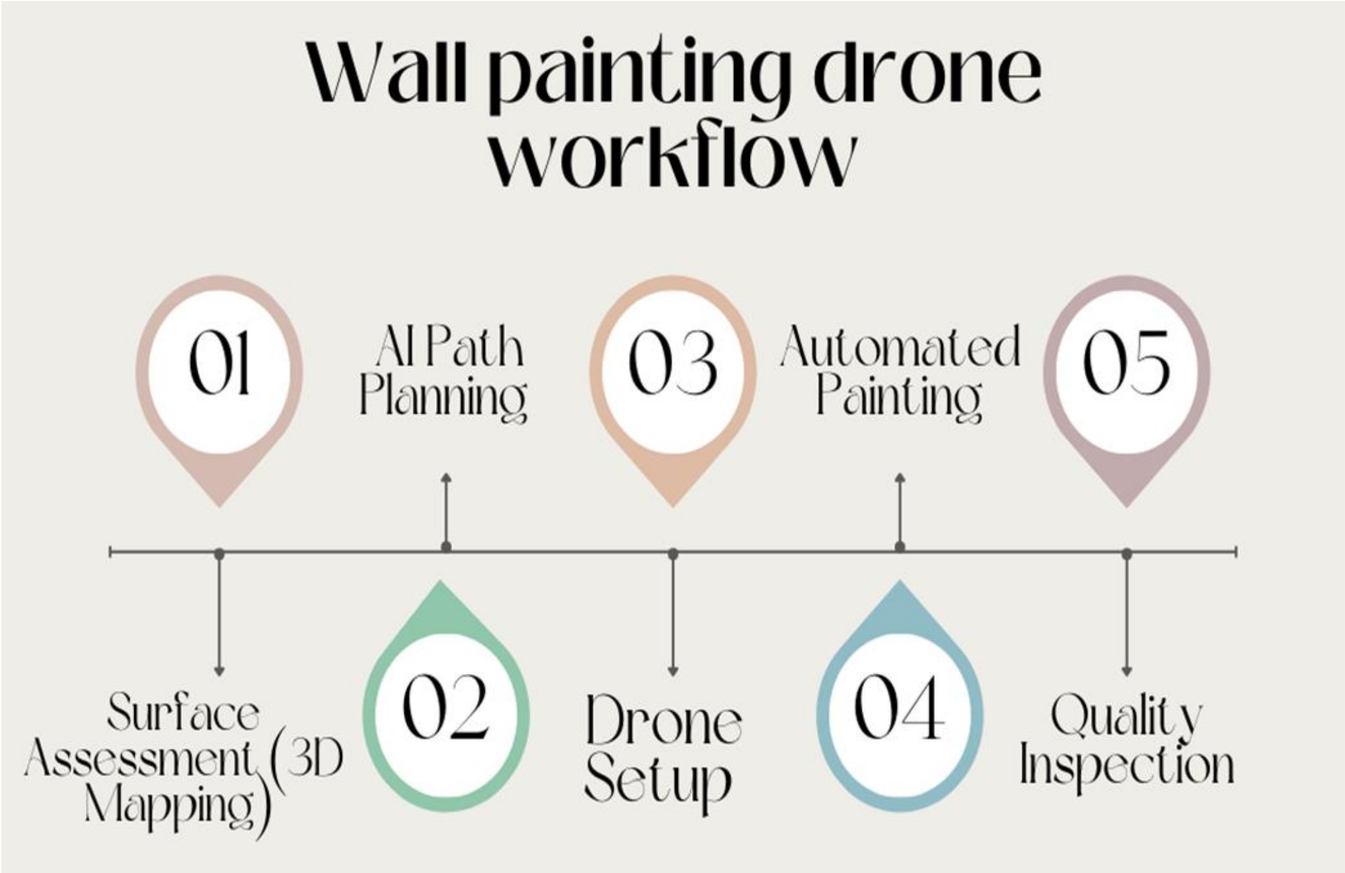
CHALLENGES IN THE PAINTING INDUSTRY

Reducing Risks, Saving Lives: Drones for Safer Wall Painting

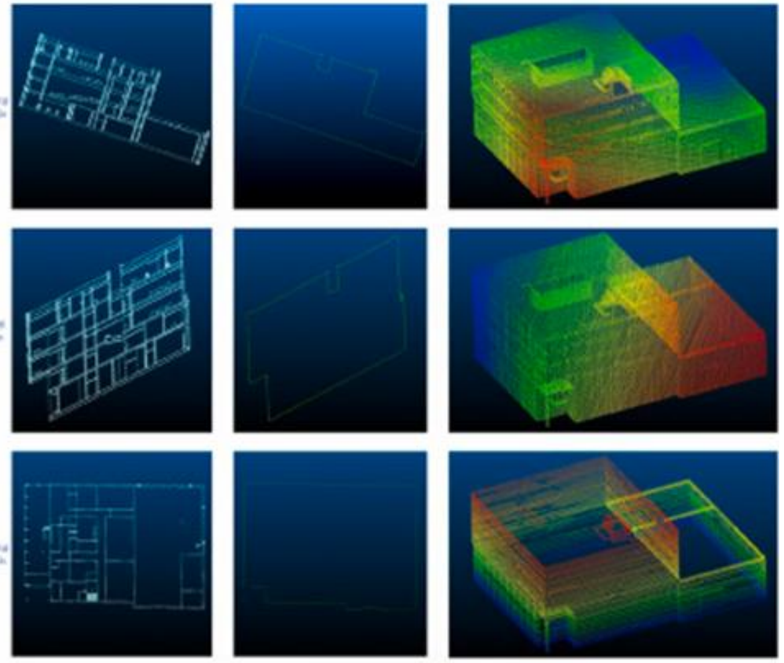
Solution: An AI-enabled autonomous drone specifically designed for painting tall and large surfaces efficiently and safely.

3D MAPPING

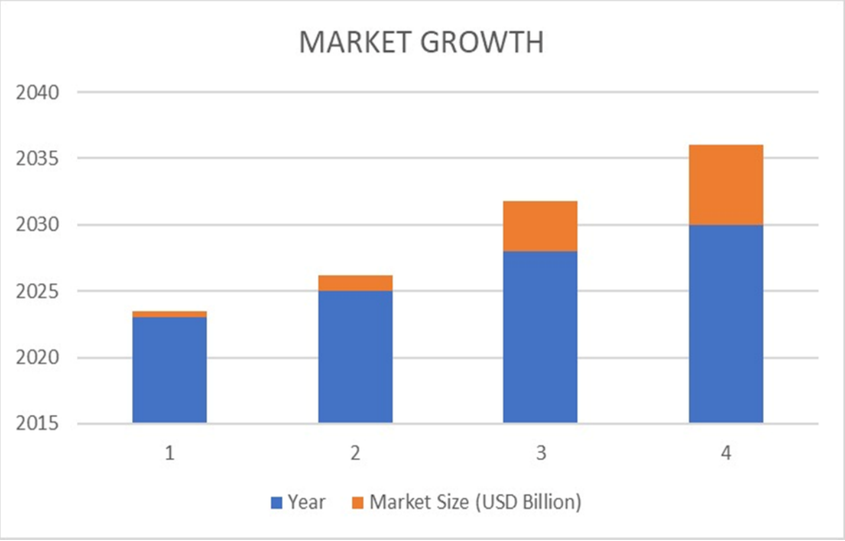




- Drones with similar capabilities (e.g., precision flying, payload delivery) already exist and are commercially available.

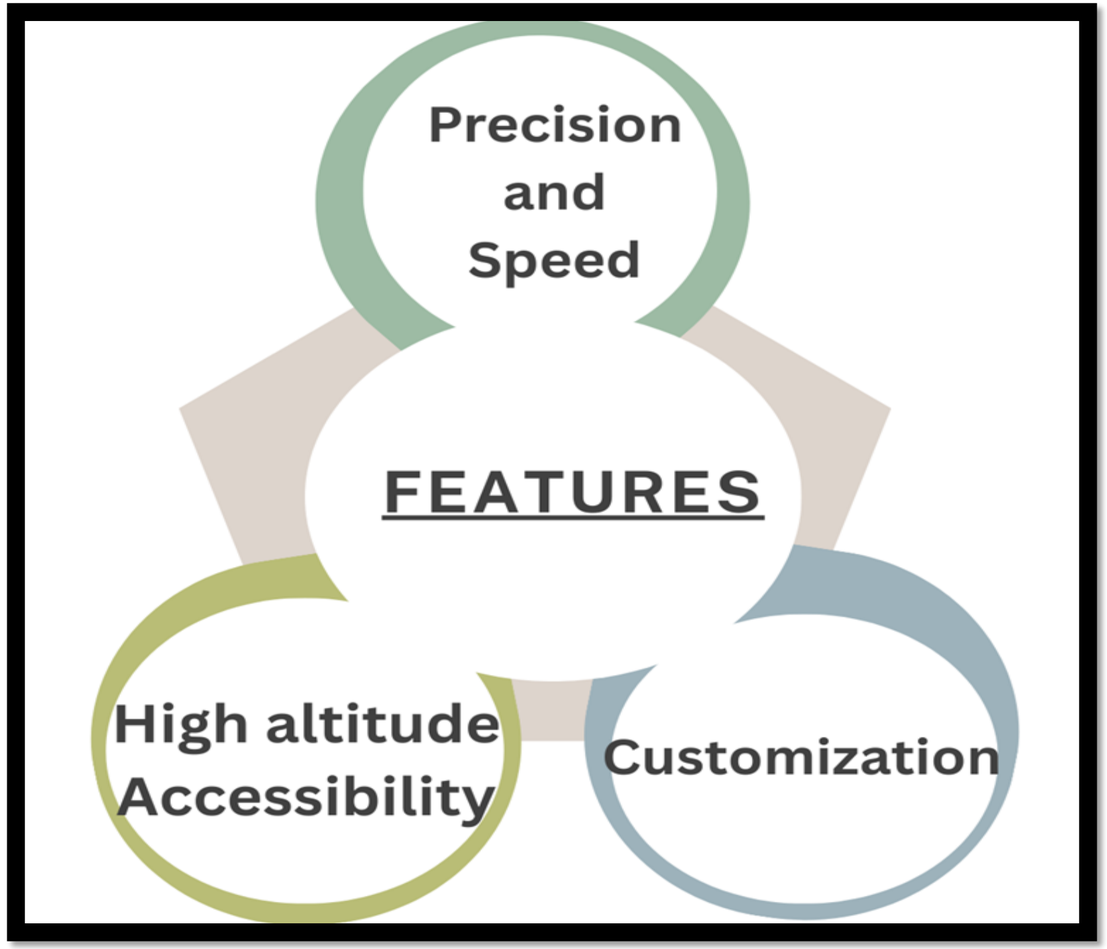


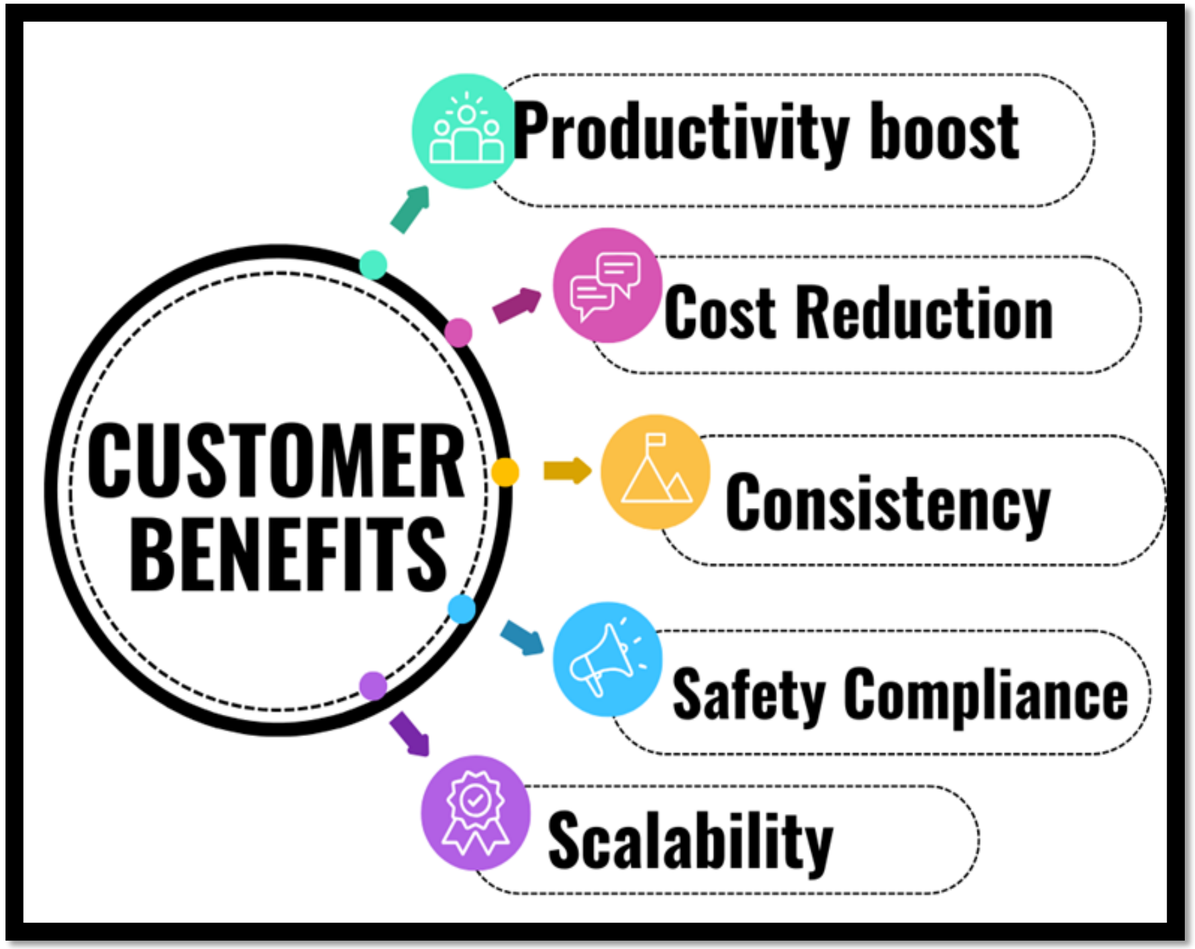
- Advanced simulation allows for **precise modeling of drone behavior**, **paint application**, and **wall navigation.**



- **Feedback** from potential users supports the development of user-friendly and effective solutions.

- Current solutions require **manual labor** whereas our drone can dynamically adapt to wall sizes, shapes and textures.





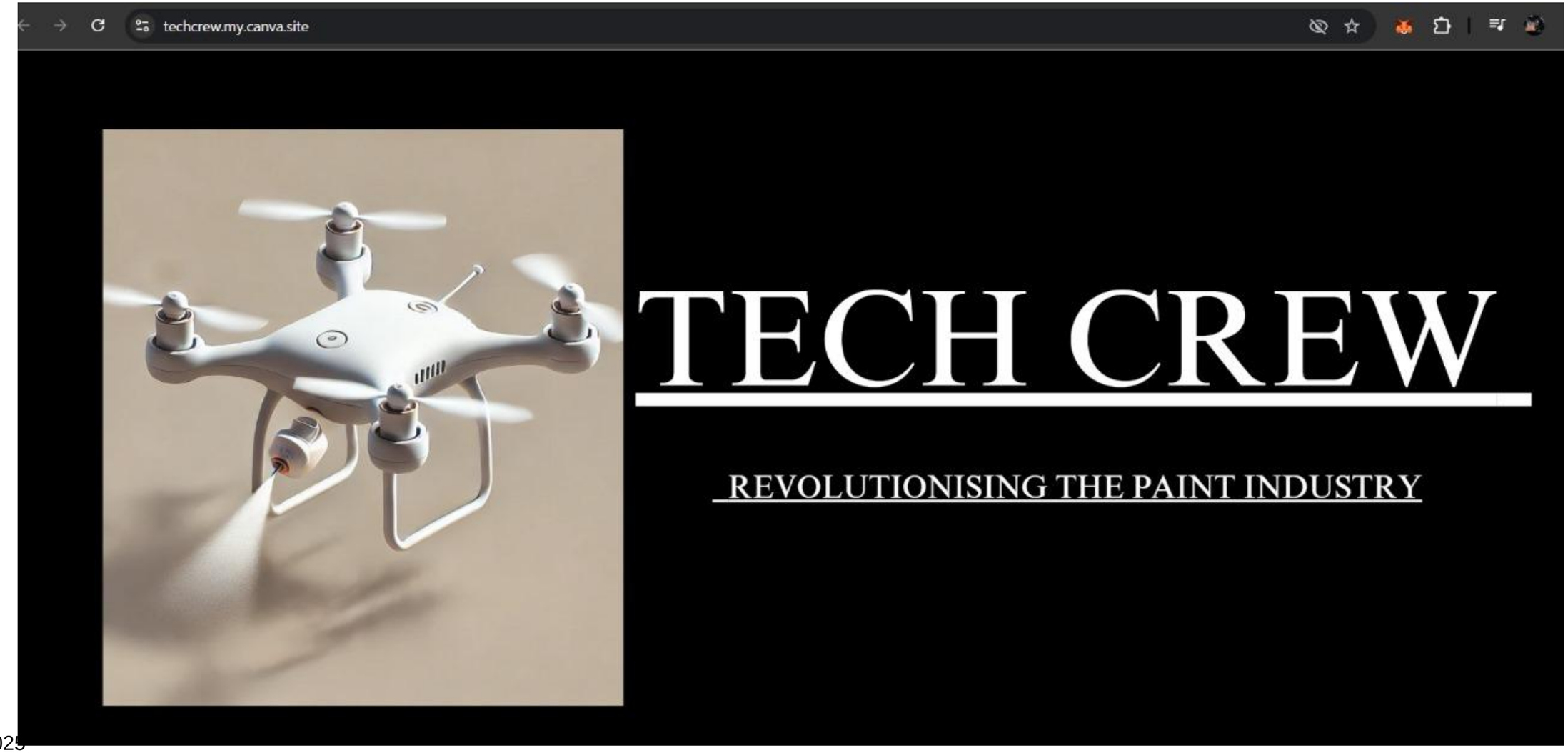
PRETOTYPE

- **Video Simulation:** Short demo of drones painting faster, safer, and better.

- **Customer Survey:** Interviews with **paint companies, contractors, and owners**.
- **Price Test:** Comparing willingness to pay for drones vs. manual painting.

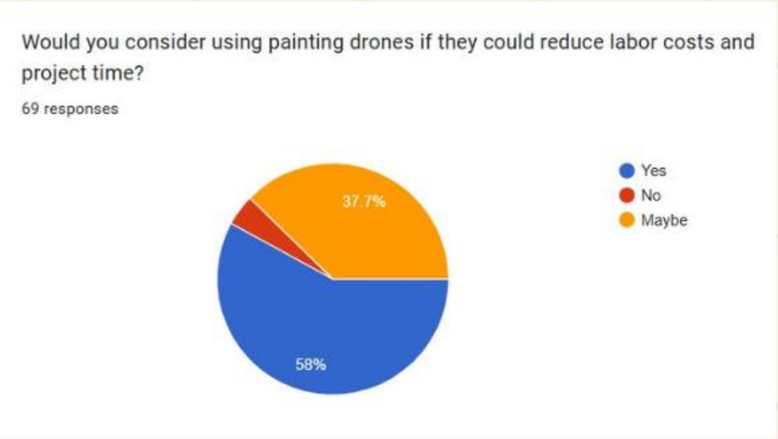
WEBSITE

OUR WEBSITE LINK-- https://techcrew.my.canva.site/

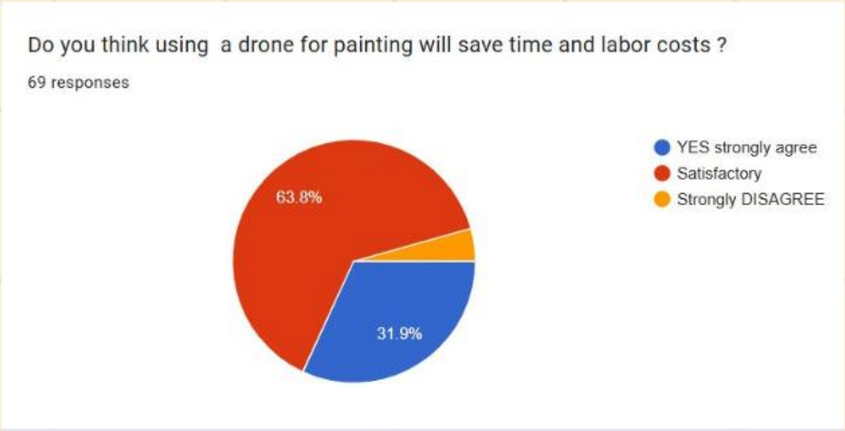


INSIGHTS FROM GOOGLE FORM

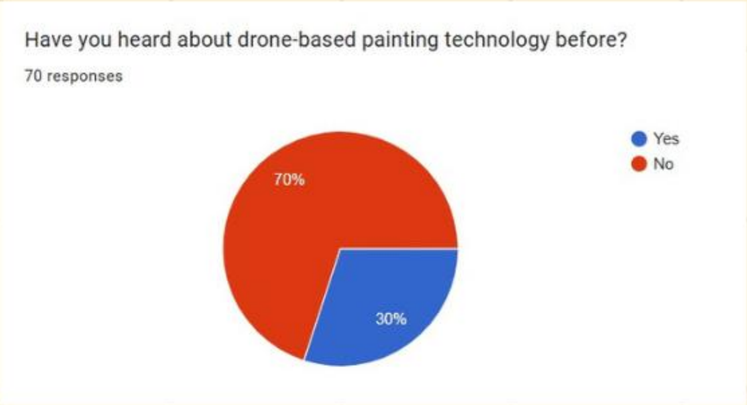
31.9% strongly agree drones save time & cost, **63.8%** find it satisfactory.

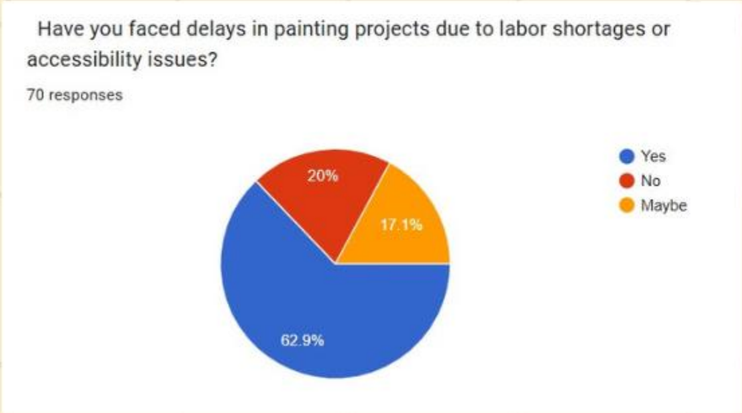


58% would use painting drones, **37.7%** are open to it.

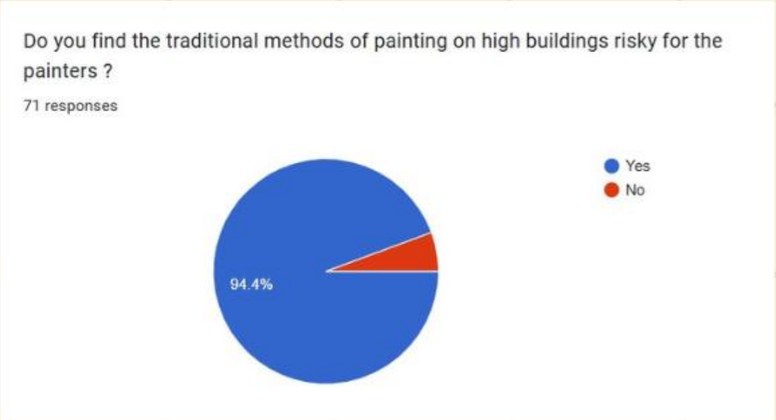


70% were unaware of drone painting tech.



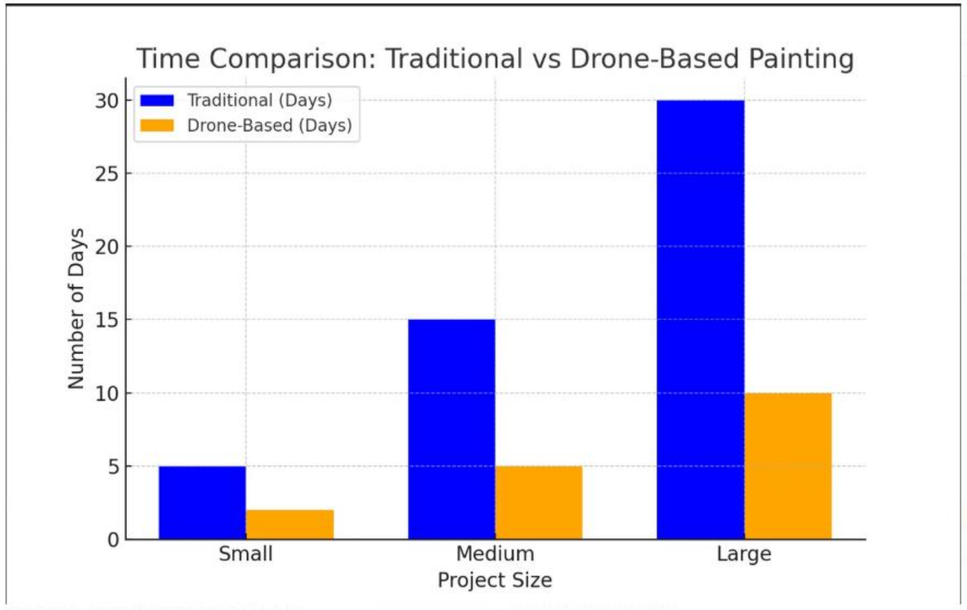


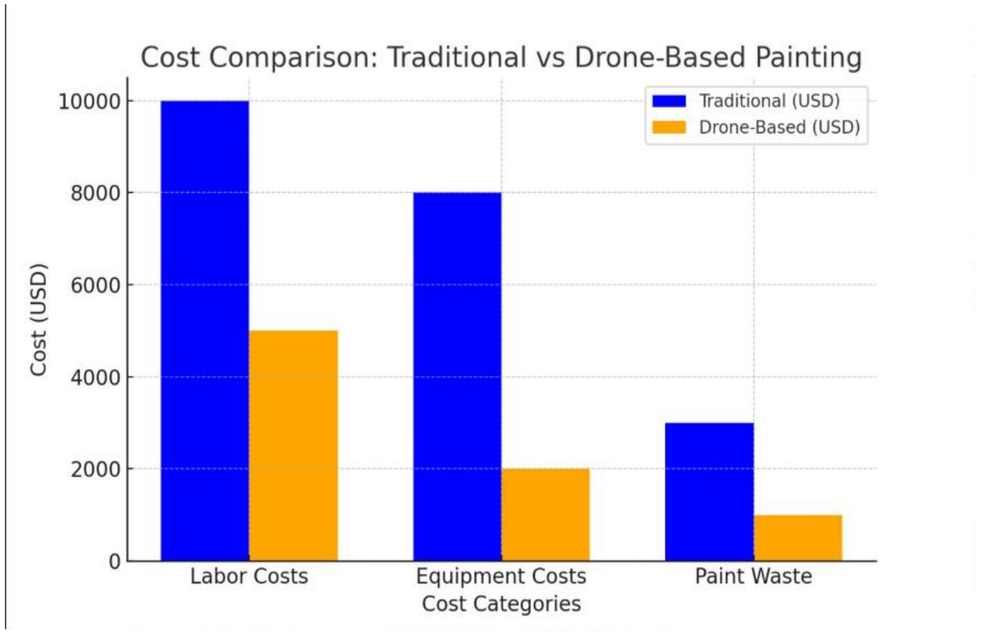
62.9% faced project delays due to labor issues.

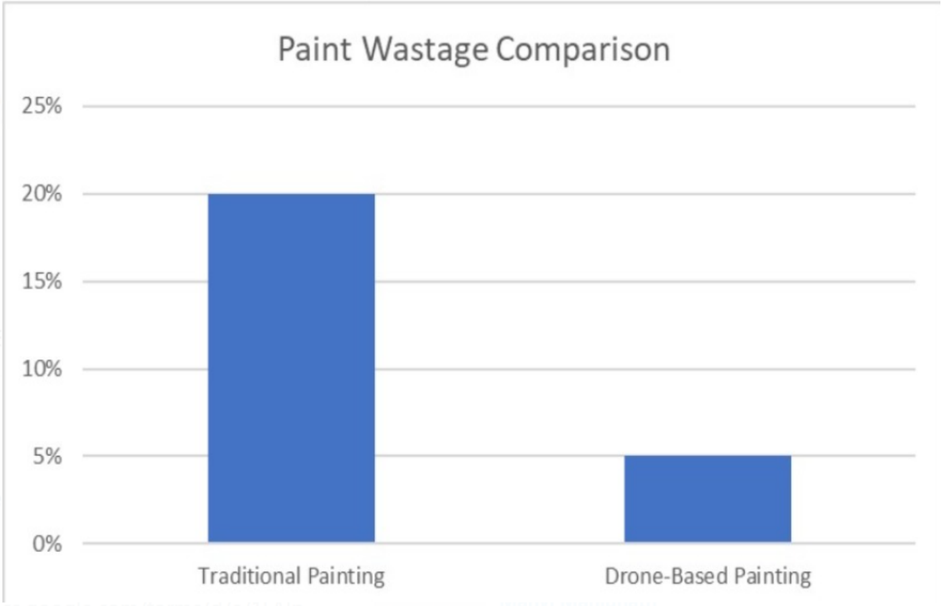


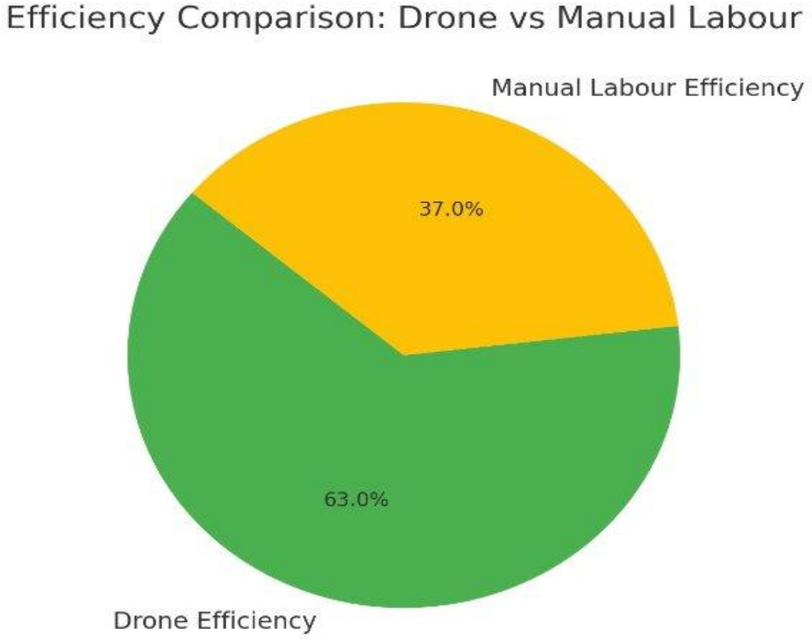
94.4% say traditional painting is risky.

Traditional Painters vs. Drone Innovations









COMPETITORS

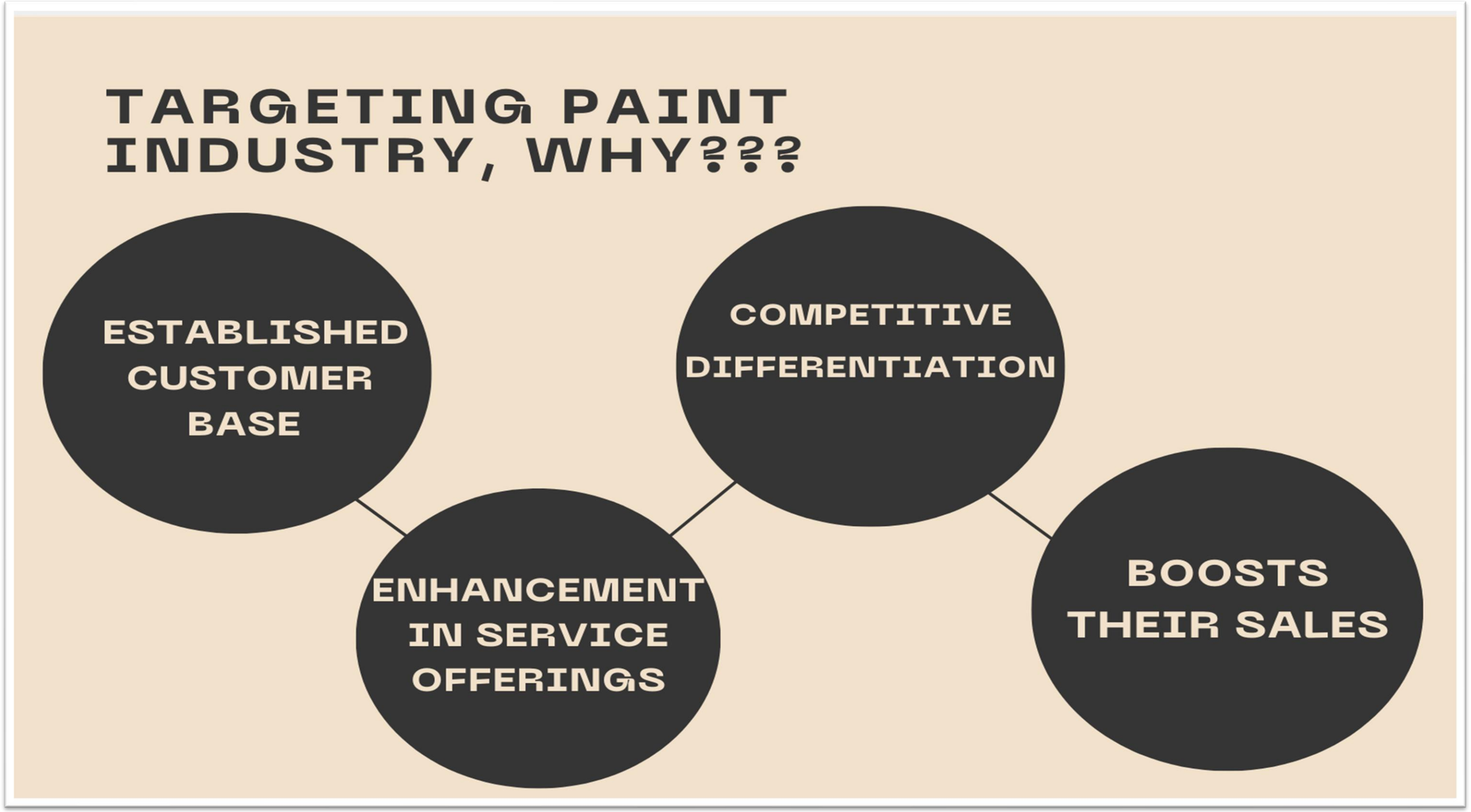
APELLIX DRONES *– No automation*

PPD painting *– No automation*

ASIAN PAINTS – Manual labor costs

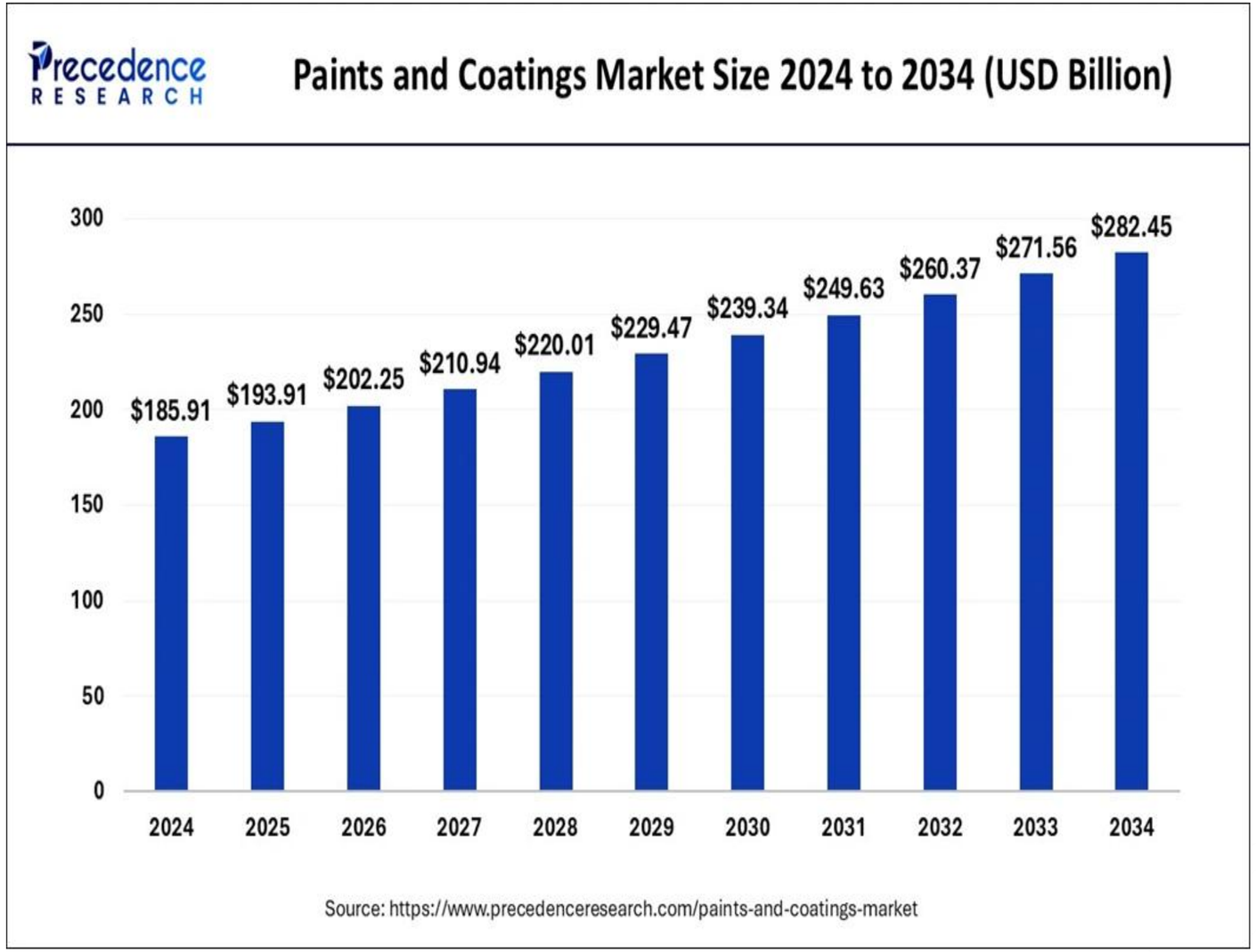
LIMITED COMPETITION!!!

Customer Discovery



Market Opportunity

- The global painting and coatings market size is estimated to be worth USD 282.45 billion by 2034.
- With our drone technology ,we aim to capture 5% of SAM(Serviceable Addressable Market) in next 3 years, equating to **14.12 million**
- Adoption of automation and AI in construction is accelerating at a CAGR of 4.27% from 2025 to 2034.



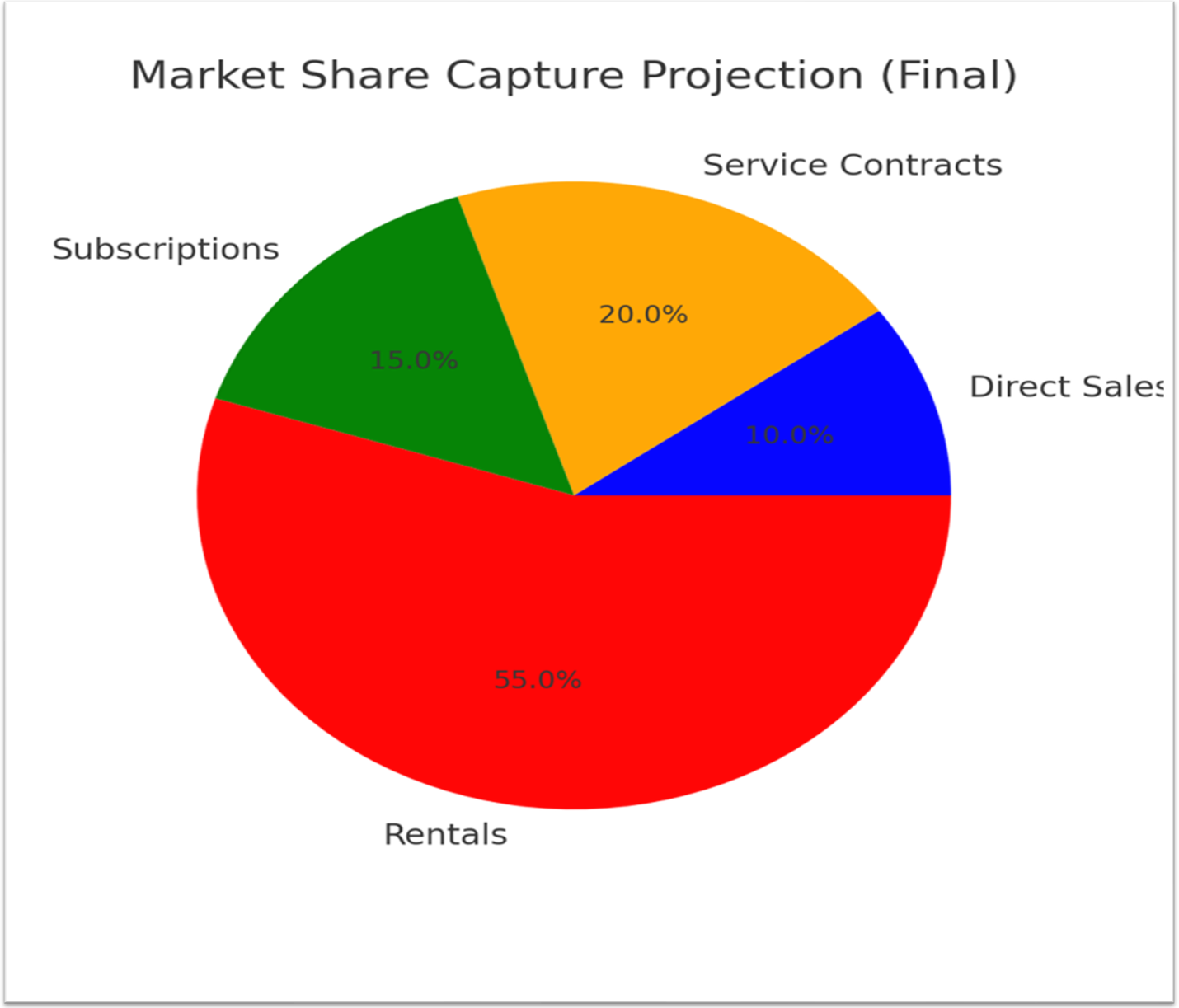
Market Catalysts

- Increasing construction activity.
- Rising demand for commercial and residential painting.
- Estimated 30% increase in automation-based solutions in construction

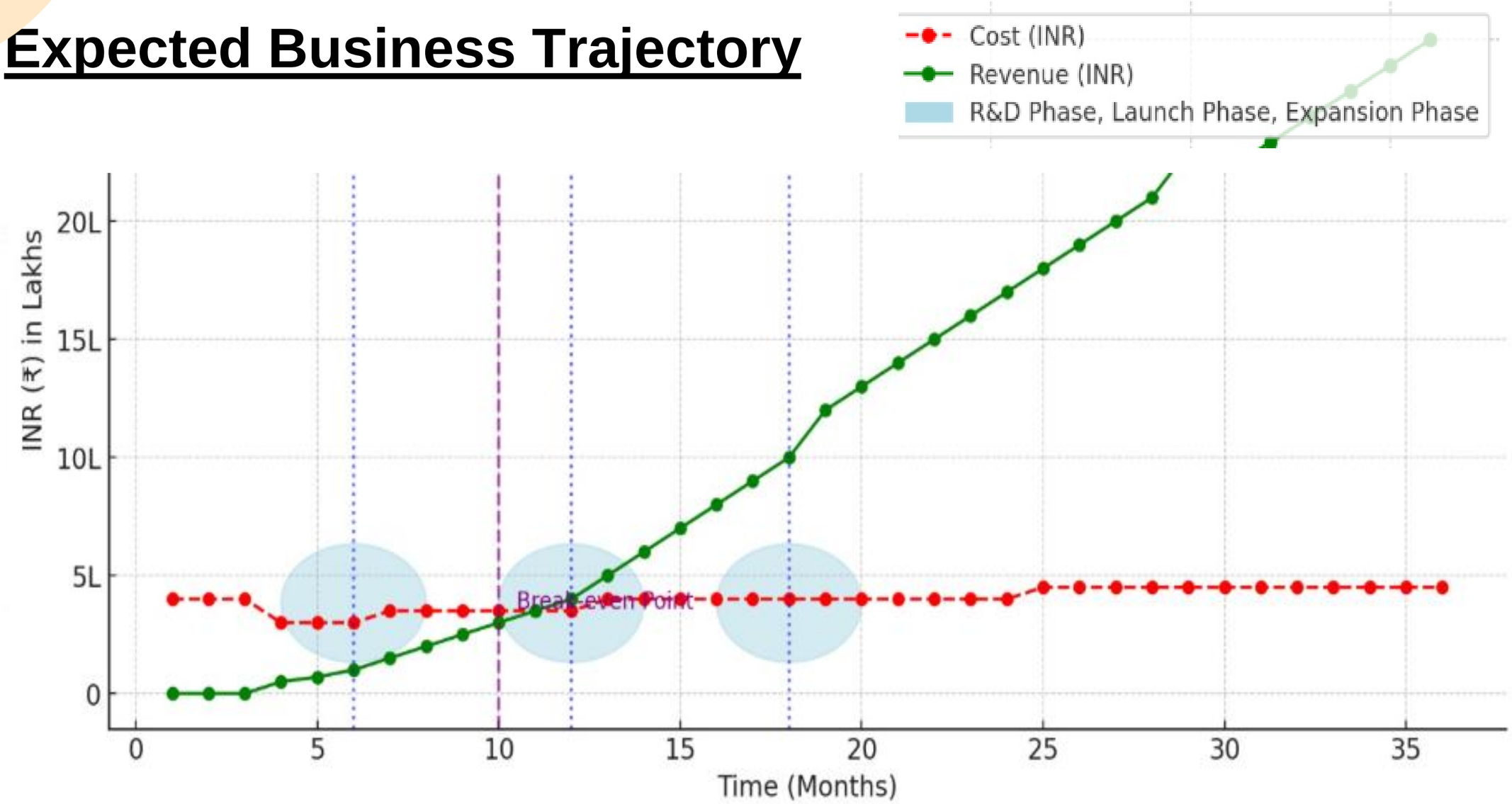
COST ESTIMATION

COMPONENT	COST
Frame (Carbon fiber for strength and stability)	8,000 – 15,000
Motors (4 to 6 high-thrust motors for stability)	10,000 – 18,000
Electronic Speed Controllers (ESCs)	4,000 – 8,000
Flight Controller (e.g., Pixhawk, Ardupilot)	10,000 – 20,000
PID Controller	Integrated with Flight Controller
Depth Camera/ LiDAR	₹15,000 – ₹25,000
Nvidia Jetson / Coral	₹12,000 – ₹15,000
Rapberry Pi -4	₹5,500 – ₹8,000
GPS Module (for positioning support)	3,000 – 5,000
RC Transmitter and Receiver (8-channel or higher)	5,000 – 10,000
Battery (LiPo 6S for extended flight time)	5,000 – 8,000
Spray System (Pump, Nozzles, Pressure Control)	15,000 – 30,000
	3,000 – 5,000
Gimbal or Camera System (for targeting precision)	5,000 – 12,000
Frame Protection (e.g., guards to prevent paint splashes)	2,000 – 5,000
Spare Parts (Propellers, Wiring, Connectors, etc.)	3,000 – 5,000
Miscellaneous (PCB design, connectors, etc.)	5,000 – 10,000
Basic Prototype Model	**₹60,000 – ₹90,000**
Advanced Model with Precision Control, Stability Enhancements, and Longer Flight Time	**₹1,50,000 – ₹2,00,000**

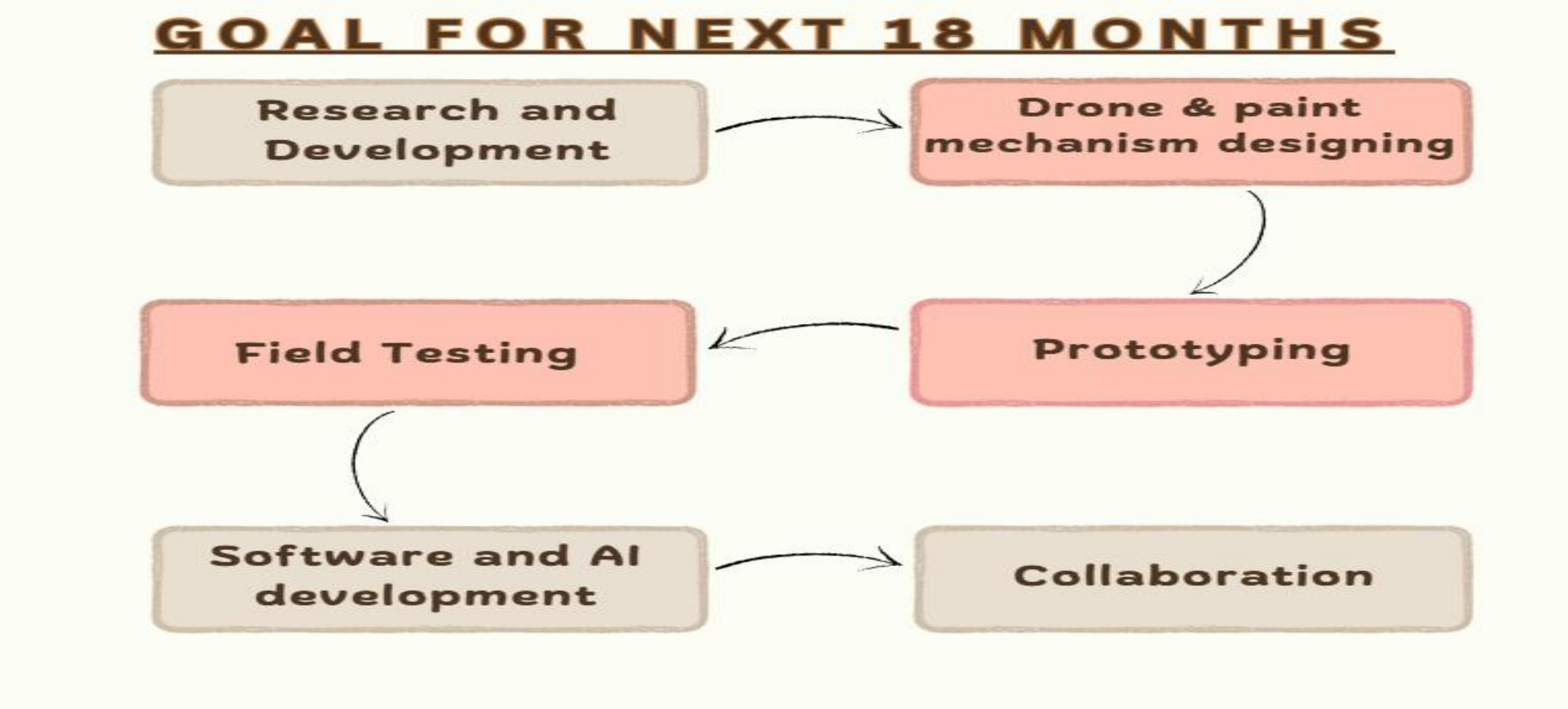
PROFIT MODEL



Expected Business Trajectory



IMPLEMENTATION PLAN



Why should investors invest in our idea?

- Construction and Real-Estate Boom
- Limited Competition
- Risk Elimination

TECH STACK

AI-Automated Painting Drone Tech Stack

1. Hardware Components

- **Drone Frame & Propulsion System**
- **Controller (e.g., Jetson Nano, ESP32, Raspberry Pi)**
- **Sensors (e.g., IMU, LiDAR, Depth Camera, Ultrasonic Sensors)**
- **Spray Painting Mechanism**
- **Communication & Control System**

2. Software Stack

- **AI Frameworks**: TensorFlow, PyTorch
- **Drone Control & Navigation**: PX4 Autopilot, ArduPilot
- **Computer Vision Algorithms**: Object Detection, Edge Detection, Image Segmentation
- **Path Planning Algorithms**: A*, Dijkstra's Algorithm, RRT (Rapidly-exploring Random Tree)
- **Cloud & Remote Management**: AWS IoT, Azure IoT Hub
- **Communication Protocols**: MQTT, ROS (Robot Operating System)

3. AI Features for Efficiency

- **Dynamic Spray Control for Uniform Coating**
- **Paint Quantity Prediction for Efficient Usage**
- **Auto-Calibration for Accurate Paint Flow**
- **Error Detection with Real-Time Feedback**

4. Recommended Architecture

- **Jetson Nano** for AI computation and image processing
- **ESP32** for lightweight control and communication
- **Raspberry Pi** for additional processing and interfacing

5. Development Tools

- **Gazebo** (Simulation Environment)
- **AirSim** (AI Training and Testing Platform)
- **QGroundControl** (Drone Flight Control Interface)
- **Python, C++** for programming and algorithm development

Business Model Canvas

Key Partners

- Paint Companies
- Construction Companies
- Interior Designers
- Event organizers

Key Activities

- Design of drone and paint mechanism
- Rental services
- Secure and build Partnership

Key Resources

- Advanced drone hardware
- Spray mechanisms
- Navigation software

Value Proposition

- Safe and reliable
- Faster and efficient
- Precision
- Customization
- Automation

Customer Relationships

- Fast repair/replacement system for any drone failures.
- customer feedback portal

Channels

- Direct Sales
- Online Platform
- Distributors

Customer Segments

- Anyone who is interested in safe customized painting

Cost Structure

- R&D expenses.
- Manufacturing setup and equipment.
- Software updates
- Battery replacements.

Revenue Streams

- Drone sales
- Rental services
- Service contracts

RESEARCH AND REFERENCES

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